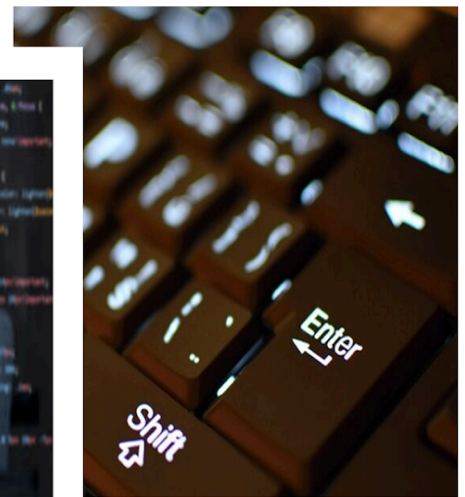


Computer Awareness

STUDY NOTES

Computer Security



Computer Security

- The definition of the term computer security has changed in recent years. Prior to the problem of data security becoming hugely publicized in the media, most people's thoughts of computer security based on the physical machine. Traditionally, computer services have been physically protected for the following three reasons:
 1. Prevention of theft of or damage to the hardware
 2. Prevention of theft of or damage to the information
 3. Prevention of disruption of service
- Computer security is safeguarding implemented to computing devices like computers and smartphones, and also in computer networks like private and public networks, including the whole Internet. The field encapsulates all the processes and mechanisms by which digital devices, information and services are safe from unintended or unauthorized access, change or destruction, and are of increasing importance in line with the increasing dependence on computer systems of the majority of societies round the world.
- It involves physical security to avoid theft of equipment, and information security to safeguard the data on that device. It is often referred to as "cyber security" or "IT security", though such terms normally do not imply physical security (locks and such).

Terminologies related to Computer Security are:

Phishing

- Phishing is the attempt to acquire sensitive information such as usernames, passwords, and credit card details (and sometimes, indirectly, money) by masquerading as a trustworthy entity in an electronic communication

Pharming

- Pharming is another technique used to steal confidential access code. It works by redirecting users to a rogue server.

SQL injection

- A SQL injection attack consists of insertion or "injection" of a SQL query via the input data from the client to the application.
- A successful SQL injection exploit can read sensitive data from the database, modify database data (Insert/Update/Delete), execute administration operations on the database (such as shutdown the DBMS), recover the content of a given file present on the DBMS file system and in some cases issue commands to the operating system.

- SQL injection attacks are a type of injection attack, in which SQL commands are injected into data-plane input in order to affect the execution of predefined SQL command,

Spyware

- **Spyware** is a type of malicious software -- or malware -- that is installed on a computing device without the end user's knowledge. It invades the device, steals sensitive information and internet usage data, and relays it to advertisers, data firms or external users.
- The capabilities include activity monitoring, collecting keystrokes, data harvesting (account information, logins, financial data), and more.
- Spyware secretly places itself on your computer and views transactions that are made on websites.

Adware

- **Adware** (short for advertising-supported software) is a type of malware that automatically delivers advertisements. Common **examples** of **adware** include pop- up ads on websites and advertisements that are displayed by software. Oftentimes software and applications offer “free” versions that come bundled with **adware**.
- Adware often appears as an advertisement and inevitably is harmful to the user.

Bot/Botnet

- A type of software application or script that performs tasks on command, allowing an attacker to take complete control remotely of an affected computer.
- A collection of these infected computers is known as a “botnet” and is controlled by the hacker or “bot-herder”
- Spyware and adware both cause a great risk dealing with credit card fraud and identification theft.

Clickjacking

- **Clickjacking** is an attack that tricks a user into clicking a webpage element which is invisible or disguised as another element. The invisible page could be a malicious page, or a legitimate page the user did not intend to visit – for **example**, a page on the user's banking site that authorizes the transfer of money.
- A hacking attack that tricks victims into clicking on an unintended link or button, usually disguised as a harmless element.

Encryption

- **Encryption** is the method by which information is converted into secret code that hides the information's true **meaning**. The science of **encrypting** and decrypting information is called cryptography. In computing, unencrypted data is also known as plaintext, and **encrypted** data is called ciphertext.
- The process of encoding data to prevent theft by ensuring the data can only be accessed with a key.
- **It helps protect private information, sensitive data, and can enhance the security of communication between client apps and servers.**

Hacker

- A hacker is a person who uses computer systems, networking or other techniques in order to overcome technical issues. The word hacker may define anyone with technical skills, but it generally refers to a person who utilises his or her abilities to attain unauthorized access to machines or networks in order to commit criminal activities.
- A hacker may, for example, steal data and info to hurt people via identity theft, damage or bring down systems and, in some cases, keep such systems hostage to demand ransom.
- The word hacker has traditionally been a divisive one, often being used as a term of admiration for a person who possesses a high degree of skill, and also creativity in his or her approach to technical issues. However, the term is generally applied to an individual who utilizes this knowledge for illegal or unethical purposes.

Types of Hackers

The security community has unofficially used definitions to hat colour as a way various hackers are identified, usually categorized into three types: white hat, black hat and grey hat.

1. White hat hackers, also called ethical hackers, strive to work in the public's best interest, instead of creating turmoil. Many white hat hackers used to perform penetration testing, hired to test to break into the company's networks to figure out and report on security vulnerabilities. The security firms then aid their clients mitigate security risks before criminal hackers may exploit them.
2. Black hat hackers by their own will acquire unauthorized access to networks and machines with malicious intentions, in order to steal data, spread malware or profit from ransom ware, vandalize or in some case

damage systems or for any other purpose, which involves gaining notoriety. Black hat hackers are clearly criminals by definition as they violate regulations against accessing machines without authorization, but they may also be involved in other illegal jobs, including identity theft and distributed denial-of-service attacks.

3. Grey hat hackers come approximately between white hat hackers and black hat hackers. While their motives can be the same as those of white hat hackers, grey hats are more likely than white hat hackers to access systems in absence of authorization; at the same time, they are more likely than black hat hackers to prevent doing intentional damage to the systems they hack. Although they aren't typically, or solely, motivated by money, grey hat hackers may help to fix vulnerabilities they have found through their own, unauthorized, activities instead of using their knowledge to exploit vulnerabilities for illegal intentions.

Vulnerability

- Vulnerability is a flaw which enables an attacker to decrease a system's information guarantee. Vulnerability is the combination of three elements: a system susceptibility or weakness, attacker permissibility to the flaw, and attacker ability to exploit the flaw. In order to exploit vulnerability, the hacker must have at least one equivalent tool or technique that can link to a system flaw. In this frame, vulnerability is also referred to as the attack surface.
- Vulnerability detection and prevention is the cyclical work of identifying, classifying, remediating, and mitigating weaknesses. This practice usually implies software vulnerabilities in computing machines.

Backdoors

- A backdoor in a computer system is a process of bypassing usual authentication, securing remote permissions to a computer, gaining access to plaintext, and so on, while trying to stay undetected.
- The backdoor can take the resemblance of an installed program (e.g., Back Orifice), or could be an amendment to an existing program or hardware component. It may also forge information about disk and memory usage.

Denial-of-Service Attack

- A **Denial-of-Service (DoS) attack** is an **attack** meant to shut down a machine or network, making it inaccessible to its intended users. **DoS attacks** accomplish this by flooding the target with traffic, or sending it information that triggers a crash.

- **DoS attack** is not to steal information but to slow or take down a web site. Denial of service attack (DOS) is an attack against a computer or network which reduces, restricts or prevents accessibility of its system resources to authorized users.

Digital Signature

- A digital signature is an electronic form of a signature that can be used to authenticate the identity of the sender of a message or the signer of a document, and also ensure that the original content of the message or document that has been sent is unchanged.

Ransomware

- Ransomware is a form of malware that encrypts a victim's files. The attacker then demands a ransom from the victim to restore access to the data upon payment.
- ***“Ransomware is extortion software that can lock your computer and then demand a ransom for its release.”***

Deepfake

- An audio or video clip that has been edited and manipulated to seem real or believable. The most dangerous consequence of the popularity of deepfakes is that they can easily convince people into believing a certain story or theory that may result in user-behavior with a bigger impact as in political or financial.

Social Engineering

- Social engineering is a manipulation technique that exploits human error to gain private information, access, or valuables.
- Scams based on social engineering are built around how people think and act. So, once a hacker understands what motivates a person's actions, they can usually retrieve exactly what they're looking for – like financial data and passwords.

Direct-Access Attacks

A malicious user gaining physical entry to a system (or part thereof) can conduct many functions, install various types of devices to compromise integrity, including operating system amendments, software worms, key loggers, and covert voice devices. The attacker may also simply download large amounts of data onto backup media, for example CD-R/DVD-R, tape; or portable drives such as key drives, digital cameras or digital audio players. One more common technique is to start an operating system stored on a CD-ROM or other bootable device and retrieve the data from the hard drive (s) this way. The only way to prevent this is to encrypt the storage device and save the key away from the system. Direct-access attacks are the only kind of threat to Standalone machines (never connect to internet), in majority of cases.

Eavesdropping

- Eavesdropping is the practice of surreptitiously monitoring a private conversation, typically between hosts connected on a network. For instance, programs like Carnivore and NarusInsight have been utilized by the FBI and NSA to eavesdrop on the machines of internet service providers.

Spoofing

- Spoofing of user identity defines a condition in which one person or program intentionally masquerades as another by falsifying information and thereby achieving an illegitimate advantage.

Snooping

- Snooping, in a security context, is unauthorized access to another person's or company's data. The practice is similar to eavesdropping but is not necessarily limited to gaining access to data during its transmission

Tampering

- Tampering defines an intentional amendment of products in the means that would turn them harmful to the user.

Repudiation

- Repudiation defines a condition where the authenticity of a signature is being compromised.

Information Disclosure

- Information Disclosure (Privacy breach or Data leak) defines a condition where information, thought as protected, is shared in an untrusted place.

Elevation of Privilege

- Elevation of Privilege defines a condition where a user or a program tries to attain elevated privileges or permit to resources that are usually restricted to him/it.

Exploits

- An exploit is a fragment of software, a chunk of data, or queue of commands that uses a software "bug" or "glitch" to cause unintended or unanticipated working to occur on computer software, machine itself, or something electronic (usually computerized). This frequently involves such things as attaining control of a computer machine or permitting privilege escalation or a denial of service attack.

The term "exploit" normally refers to small programs created to take advantage of a software weakness that has been revealed, either remote or local. The code from the exploit program is periodically reused in Trojan horses and computer viruses.

Indirect Attacks

- An indirect attack is an attack conducted by a third-party computer machine. By using a second person's computer to conduct an attack, it becomes far more challenging to track down the real attacker. There have also been events where attackers used the anonymity of public systems, such as the tor onion router system.

Computer Security Types

One way to ascertain the similarities and variations among Computer Security is by enquiring what is being secured. For example,

1. Information security is protecting information from illicit access, modification & deletion
2. Application Security is protecting an application by constructing security features to restrict from Cyber Threats just like SQL injection, DoS attacks, data breaches, etc.
3. Computer Security defines protecting a standalone system by keeping it updated and patched
4. Network Security is by protecting both the software and hardware technologies
5. Cyber security is referred as protecting computer machines, which are linked over the computer networks

Types of Attacks

Attacks are categorized into two types:

1. **Passive attacks:** does not include any amendments to the body of an original message
2. **Active attacks:** the body of the original message are amended in some ways.

Components of Computer System

The devices of a computer system that required to be protected are:

1. Hardware, the physical part of the machine, like the system memory and disk drive
2. Firmware, permanent software that is written into a hardware device's non-volatile memory and is in reality invisible to the user
3. Software, the programming that facilitates services, like operating system, word processor, net browser to the user

The CIA Triad

Computer security is primarily concerned with three main categories:

1. Confidentiality is assuring that information is provided only to the intended customers
2. Integrity is securing information from being amended by unauthorized personals
3. Availability is that information must be provided to users as and when required.



In easy language, computer security is assuring that information and computer components are working but still secured from people or software that shouldn't get access to it or modify it.

Computer Security Threats

- Computer security threats are possible problems that can possibly hamper the usual working of the computer system. In the current times, cyber threats are exponentially increasing as the world is going digital. The very common harmful types of computer security are follows:

Trojans

- A Trojan horse, or Trojan, is a type of malicious code or software that looks legitimate but can take control of your computer.
- A Trojan is designed to damage, disrupt, steal, or in general inflict some other harmful action on your data or network.

Viruses

- A computer virus is a malicious program which is transmitted into the user's computer without the user's knowledge. It copies itself and affects the files and programs on the user's PC. The actual intention of a virus is to make sure that the victim's computer will never be able to work properly or even at all.



Computer Worm

- A computer worm is a kind software program that can replicate itself from one computer to another, without manual involvement. The potential vulnerability here is that it will utilize up user's computer hard disk space as a worm can copy in large volume and with very high speed.



Phishing

- Disguising as an authentic individual or business, phishers try to steal sensitive financial or personal data through fraudulent email or instant mails. Phishing is in real unfortunately, very simple to execute. The user is deluded into thinking it's genuine mail and he/ she may enter his/her personal information.



Botnet

- A botnet is a cluster of computers linked to the internet that have been compromised by a hacker through a computer virus. This particular computer is known as a 'zombie computer'.



Rootkit

- A rootkit is a computer program created to facilitate continued privileged permissions to a computer while actively hiding its activity. Once a rootkit has been loaded, the controller of the rootkit will have the ability to remotely control and execute files and change system specifications on the host machine.



Key logger

- Also called a keystroke logger, key loggers can track the real-time working of a user on his/her computer. It maintains a record of all the keystrokes done by user keyboard. Key logger is also a very strong threat to steal people's login credentials, namely, username and password.
- These are perhaps the very common security threats that anyone can come across. Apart from these, there are different ones like

spyware, wabbits, scareware, bluesnarfing and so on. Fortunately, there are techniques to secure the system against these attacks.



Logic Bomb

- A logic bomb is a piece of code intentionally inserted into a software system that will set off a malicious function when specified conditions are met. For example, a programmer may hide a piece of code that starts deleting files (such as a salary database trigger), should they ever be terminated from the company.
- Software that is inherently malicious, such as viruses and worms, often contain logic bombs that execute a certain payload at a pre-defined time or when some other condition is met.
- This technique can be used by a virus or worm to gain momentum and spread before being noticed. Some viruses attack their host systems on specific dates, such as Friday the 13th or April Fools' Day. Trojans and other computer viruses that activate on certain dates are often called "time bombs".

Why is Computer Security Important?

- In this digital age, we all want to protect our computers and our personal data and hence computer security is crucial to keep our personal information protected. It is also crucial to maintain our computer protection and its overall health by prohibiting viruses and malware which would affect the system performance.

What is Anti-Virus?

- Antivirus software is a commonly available and widely used program or set of programs that are designed to restrict, search for, detect, and remove software viruses, and other malicious software like worms, Trojans, adware, and more.
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Why Do We Need Antivirus Software?

- These tools are crucial for users to have loaded and up-to-date because a computer in absence of antivirus software protection will be infected within minutes of connecting to the internet. The bombardment is constant, which implies antivirus companies have to update their detection tools periodically to tackle the more than 60,000 new pieces of malware created every day.
 - Today's malware (an umbrella term that encapsulates computer viruses) alters appearance suddenly to prevent detection by older, definition-based antivirus programs. Viruses can be designed to cause damage to the device, prohibit a user from accessing data, or to take control of the computer.
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What Does Antivirus Software Do?

- Various companies create antivirus software and what everyone of them offer can be different but all perform following essential functions:
 1. Scan particular files or directories for any malware or predetermined malicious patterns
 2. Allow user to plan scans to automatically run for user

3. Allow users to conduct a scan of a selected file or the entire computer, or of a CD or flash drive at any time.
 4. Quarantine any malicious piece of program detected, sometimes the user will be notified of such an infection and asked if he/she wants to dump the file, other programs will automatically do this in the background.
 5. Reminds about the 'health' of the computer
- Always be sure the user has the best, up-to-date security software installed to protect his/her devices like computers, laptops, tablets, and smartphones.

How Does Antivirus Software Work?

Most of the antivirus software programs still download malware definitions straight to the user's device and scan the files in search of any possible matches. But since, as it's mentioned above, most malware periodically morphs in appearance to avoid being caught, Web root works differently. Rather than storing examples of recognized malware on the device, it saves malware definitions in the cloud storage. This enables us to acquire less space, scan quicker, and manage a more robust threat library.

Free vs. Paid Antivirus Software

- Activities from banking to baby photos, most of our business and personal data stays on our devices. If it were saved physically, spending for a security application would be a no-brainer.
- Unfortunately, we usually expect our online data to stay protected without lifting a finger or spending anything. Companies encouraged to do it without any cost are partly responsible for the ambiguity, to be sure.
- But consumers must insist on features like identity theft security, mobile security, and support facilities when it comes to their data security, too—features normally lacking with priceless solutions.

What is Firewall?

- A firewall is a kind of security-conscious type of hardware or software that stays between the computer and our network with a primary task: preventing malicious software from reaching us. In other words we can say that the firewall works like a security guard between the Internet and our local area network (LAN). All network traffic passing through the LAN must face the firewall, which restricts unauthorized access to the network.

Types of Firewalls

1. **Packet-filtering firewall:** It checks every packet that passes the firewall and examines the packet as per the set of rules that the user sets up. If the packet gets to pass the test, it's allowed to go through. If the packet fails to pass the test then it is rejected. Packet filters are the most inexpensive variety of firewall. That is why packet-filtering firewalls are used very commonly.
2. **Stateful packet inspection (SPI):** It is installed in accordance with simple packet filtering. A firewall having Stateful packet checking looks at packets in clusters instead of doing individually. It maintains the record of all those packets that have crossed through the firewall and can create patterns that indicate unauthorized entry.
3. **Deep packet inspection (DPI):** It is the actual data of the packets that is checked so viruses, spam, and other harmful content can be restricted. DPI also enables practices, such as data mining, eavesdropping, and content censorship, which make its way through a controversial subject.

Few Cyber Crime Prevention Techniques

1. **Use Strong Passwords:** Using different **user ID / password combinations** for various accounts and not writing them down. Making the passwords more **complex** by using letters, numbers, special characters (minimum 10 characters in total) and changing them periodically.

2. Securing computer:

- a. **Activate the firewall:** Firewalls are the **primary line of cyber defence**. They **restrict connections** to unknown or bogus websites and will keep out various types of viruses and hackers.
 - b. **Use anti-virus/malware software:** Prohibit viruses from infecting the computer by **loading and periodically updating anti-virus software**.
 - c. **Block spyware attacks:** restrict spyware from infiltrating the computer system by installing and updating anti-spyware programs.
3. **Being Social-Media Savvy:** Making sure that social networking profiles (e.g. Facebook, Twitter, YouTube, MSN, etc.) are made private. Checking the **security settings**. Being careful what information is being posted online. Once it is on the Internet, it stays there forever!
 4. **Securing Mobile Devices:** Being aware that the mobile device is **vulnerable to viruses and hackers**. Downloading applications from **trusted sources**.
 5. **Install the latest operating system updates:** Keeping the applications and operating system (e.g. Windows, Mac, Linux, etc.) current with the earliest **system updates**. Turning on automatic updates to **prohibit potential attacks** on older software.
 6. **Protecting the Data:** Using **encryption** for the most sensitive documents such as **tax returns or financial records**, making regular back-ups of all the important data, and storing it in **different locations**.
 8. **Securing the wireless network:** Wi-Fi (wireless) networks at **residence** are vulnerable to intrusion if they are not correctly **protected**. **Reviewing and modifying** default settings periodically. Public Wi-Fi, also known as “**Hot Spots**”, are also **vulnerable**. Avoiding **conducting financial or corporate transactions** on these networks.
 9. **Protecting the e-identity:** being very vigilant when **sharing personal information** such as name, address, phone number or financial information

on the **websites or emails**. Making sure that websites are **protected** (e.g. when making online purchases) or that **privacy settings** are enabled (e.g. when accessing/using social networking sites).

10. **Avoiding being scammed:** Always **checking before clicking** on a **link or file of unknown origin**. Not feeling compelled by emails. Checking the **source of the message for authenticity**. When doubtful, verifying the source. Avoiding replying to emails that ask to verify the information or confirm the user ID or password.
11. **Calling the right person for help:** Not getting panicked! If anyone is a victim, if encountered illegal **Internet content** (e.g. child exploitation) or if suspecting a computer crime, identity theft or a commercial scam, reporting this to the local law authority. If need help with maintenance or software installation on the computer, consulting with an authorised service provider or a certified computer technician.